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Predictive Analytics for Early Disease Detection

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Abstract

Early detection of life-threatening conditions like Sepsis, Heart Failure, and Chronic Kidney Disease (CKD) in ICU patients can significantly improve patient outcomes and reduce healthcare costs. This project applies interpretable machine learning (ML) models to 24-hour pre-ICU data from the MIMIC-IV database to predict these diseases. We construct a clinical snapshot using structured vitals, labs, scores, and interventions, and apply Logistic Regression, Random Forest, Gradient Boosting, and XGBoost models. Our experiments evaluate model performance on both balanced and imbalanced datasets. Results indicate that Random Forest and Gradient Boosting models offer the best trade-offs between recall and interpretability, crucial for deployment in clinical settings. The study emphasizes domain-informed feature selection, real-world data preprocessing, and clinically sensitive evaluation strategies.

Introduction

In critical care environments like the Intensive Care Unit (ICU), patients often arrive in rapidly deteriorating condition. Conditions such as Sepsis, Heart Failure, and Chronic Kidney Disease (CKD) are especially common and dangerous, and early detection of these conditions can greatly improve a patient's chances of recovery. However, in practice, detecting these diseases early especially before a patient is officially admitted to the ICU is extremely challenging. Clinicians often rely on standardized scoring systems like SOFA (Sequential Organ Failure Assessment) and qSOFA (quick Sepsis related Organ Failure Assessment) to assess patient severity. These tools are widely used and clinically validated, but they are typically applied after ICU admission and are based on a limited number of variables with fixed thresholds. As a result, they can miss more subtle, earlier signs of deterioration. What's missing in current workflows is a way to predict, with clarity and confidence, who is at risk before they reach that critical stage. This project addresses that gap. We use real patient data from the MIMIC-IV clinical database to explore whether machine learning models can detect early signs of Sepsis, Heart Failure, and CKD by analyzing just the 24-hour window leading up to ICU admission. MIMIC-IV contains rich, de-identified EHR data from tens of thousands of ICU stays, including vitals, lab results, medications, and diagnoses making it an ideal dataset

for building a predictive system grounded in clinical reality. Rather than including every available feature, we carefully selected 41 clinically relevant features based on medical domain knowledge. These include vitals like heart rate and respiratory rate, lab test results such as creatinine and BUN, and intervention flags for treatments like vasopressors or antibiotics. This selection process ensured that the models would rely on inputs that are both available and meaningful in a real-world hospital setting. We trained separate models for each disease Sepsis, Heart Failure, and CKD using machine learning algorithms such as Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. The models were evaluated on both balanced and real-world imbalanced datasets to assess their effectiveness under different clinical conditions. Our focus throughout was on achieving a balance between recall and precision. In a clinical setting, high recall helps ensure that fewer positive cases are missed, which is essential for patient safety, while high precision helps avoid false alarms and ensures that clinicians receive reliable alerts they can act on. Importantly, our goal was not to build a black-box system. Instead of relying on complex deep learning methods, we chose models that are commonly used in clinical ML applications and are known for their robustness, interpretability, and performance on structured tabular data. This aligns with the growing need in healthcare for AI systems that are transparent, easy to validate, and trustworthy by clinicians. This work is not about replacing clinical judgment. Rather, it's about building tools that could support ICU staff by providing early alerts based on patterns that are difficult to catch manually. With models grounded in clinical features and supported by real-world data, we aim to take a step toward more proactive, data-driven ICU care.

Literature Review

The integration of machine learning (ML) techniques into healthcare has shown great promise, particularly for early detection of critical conditions in Intensive Care Units (ICUs). In the context of ICU patient monitoring, early identification of conditions such as Sepsis, Heart Failure, and Chronic Kidney Disease (CKD) can significantly improve clinical outcomes. However, traditional methods, such as the SOFA score [1], are primarily used reactively and are limited by their reliance on fixed thresholds. While the SOFA score is effective for monitoring organ failure during ICU

stays, it lacks the ability to predict deterioration before ICU admission, thereby missing opportunities for early intervention. Several studies have focused on using machine learning models to predict patient outcomes before ICU admission. Harutyunyan et al. [2] demonstrated the potential of multitask learning using MIMIC-III data to predict mortality and other outcomes. While these models achieved promising results, their lack of interpretability poses a significant challenge for clinical adoption. Medical professionals need to trust the predictions made by AI systems, and this trust can only be established when the models are interpretable and provide clear insight into how decisions are made. A critical review by Shickel et al. [3] emphasized the importance of interpretability in healthcare-related machine learning. The authors pointed out that while deep learning models can provide high accuracy, their black-box nature makes them unsuitable for direct clinical use. Tree-based models, such as Random Forests and Gradient Boosting Machines, have become popular for clinical applications due to their ability to deliver high performance without sacrificing transparency. These models allow clinicians to gain insights into which clinical features are most predictive of patient outcomes, making them valuable tools in healthcare settings. In the context of Sepsis, Heart Failure, and CKD, several studies have utilized machine learning to identify patients at risk. Komorowski et al. [4] proposed a reinforcement learning framework to optimize treatment strategies for Sepsis in ICU patients. While this approach is promising, it requires real-time monitoring, which may not be feasible in many clinical environments. On the other hand, in the context of Sepsis, various studies have explored the impact of drugs and interventions on patient outcomes. For instance, Wiewel et al. [5] conducted a prospective observational study that investigated the prior use of calcium channel blockers in critically ill patients with Sepsis. They found that patients who had previously used calcium channel blockers had a reduced mortality rate, suggesting a possible protective effect. This study aligns with our focus on Sepsis prediction, as it highlights the importance of early intervention and predictive factors in Sepsis management. For Heart Failure, prediction models typically rely on clinical history, vital signs, and laboratory results. Several studies have applied ensemble methods, such as Random Forests and Gradient Boosting, to predict Heart Failure outcomes. These methods have demonstrated strong performance. For example, Random Forests have been particularly effective in identifying at-risk patients due to their ability to handle complex data interactions and their interpretability, which is crucial for clinical adoption. These models can help identify patients at risk for acute exacerbations, allowing for earlier interventions that may improve patient outcomes. Similarly, in Chronic Kidney Disease (CKD) prediction, machine learning models have used features such as creatinine levels, eGFR, and BUN to identify early signs of kidney dysfunction. Previous work has shown that combining these features with machine learning can overcome some of the challenges associated with delayed or incomplete data, enabling the identification of patients at risk before they experience severe complications. This work differs from existing studies by focusing

on a 24-hour pre-ICU window. Rather than relying on real-time monitoring or deep learning techniques, which may not always be feasible or interpretable, our project uses interpretable models like Random Forest and Gradient Boosting, ensuring that the outputs are both accurate and understandable for clinicians. By using clinically relevant features and applying machine learning to structured data, we aim to create a predictive tool that can be seamlessly integrated into clinical workflows to support early detection and intervention. In conclusion, while existing literature has focused on applying machine learning in ICU settings to predict Sepsis, Heart Failure, and CKD, most of these approaches rely on continuous data monitoring or complex models that may not be practical for all settings. By focusing on a 24-hour pre-ICU prediction window and using clinically relevant features, this work aims to develop a practical, data-driven tool to assist in early detection and intervention in the ICU. The methodology section describes the key steps followed in the project, including data preparation, feature selection, and the machine learning pipeline setup for predicting Sepsis, Heart Failure, and Chronic Kidney Disease (CKD). We utilized the MIMIC-IV dataset, containing clinical data of ICU patients, to extract relevant features, preprocess the data, and implement various models.

Methodology

1. **Data Preparation:** The MIMIC-IV v2.2 dataset was used for this project [6], obtained through PhysioNet, a well-established repository for healthcare data. Due to the sensitive nature of clinical data, access to MIMIC-IV required completion of the CITI Program's [8] "Data or Specimens Only Research" certification, which took approximately one month to complete. This certification ensured that we understood and adhered to the ethical guidelines for handling patient data, particularly in the context of human subjects research.

This process involved the following steps:

- (a) **Loading Key MIMIC-IV Tables [7]** : Imported ICU-related clinical tables: patients, admissions, transfers, chart events, lab events, and diagnoses.
Queried over 400 million rows from chart events for real-time vitals and monitoring data.
- (b) **Building Adult ICU Cohort:**
Filtered for adult patients (≥ 18 years) using the `anchor_year` field.
Ensured ICU admissions were accurately matched using the transfers table.
Excluded short ICU stays (< 4 hours) to retain clinically relevant data.
- (c) **Finalizing ICU Stay Timelines:** Retained only the first ICU stay for each patient to reduce bias from repeated ICU visits.
Defined a 24-hour pre-ICU time window to extract early signals (e.g., vital signs, labs, interventions) to help detect early indicators of deterioration.
Integrated metadata (age, gender, etc.) and patient-specific information for cohort analysis.

	A	B	C	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO
1	patient_id	admission_age		c_count	lactate	platelet_c	bilirubin	nt_probnp	troponin	albumin	calcium	phosphate	hemoglobi	received_v	received_c	received_eiv	fluid_voi	shock_index	egfr	qsofa_sco	bun_creati	anion_gap	has_sepsis	has_heart	has_ckd	
2	10000032	29079034	52	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
3	10000690	25860671	86	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
4	10000980	26913865	73	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	50	0.6875	65.98807	0	18.66667	16	0	0	
5	10001217	24597018	55	15.7	1.7	283	0.6	4	4	3.6	7.9	3.6	12.5	0	0	0	0	0	0.6875	103.7915	0	16.66667	16.2	0	0	
6	10001217	27703517	55	5.4	1.7	301	0.4	4	4	3.6	9	3.8	12.5	0	0	0	0	0	0.6875	128.0964	0	22	12.1	0	0	
7	10001725	25563031	46	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
8	10001843	26133978	73	13.8	2.4	551	4.2	2	2	2.4	8.9	5.2	9.5	0	0	0	0	0	0.6875	54.11191	0	21.53846	15.3	1	0	
9	10001884	26184834	68	12.2	1.7	218	0.6	20	20	3.6	10.2	3	13	0	0	0	0	0	0.6875	49.39415	0	24.54545	16.5	0	0	
10	10002013	23581541	53	9.5	2.3	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
11	10002155	20345487	80	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
12	10002155	28994087	80	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
13	10002348	22725460	77	11.8	1.7	252	0.6	4	4	3.6	8.8	3.1	12	0	0	0	0	1000	0.627273	69.55342	1	37.5	13	0	0	
14	10002428	20321825	80	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
15	10002428	23473524	80	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
16	10002428	28662225	80	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
17	10002667	23197839	58	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
18	10002760	28094813	56	9.5	1.4	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
19	10002930	25696644	48	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
20	10002930	25922998	48	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
21	10003019	22774359	69	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.740741	65.98807	1	18.66667	16	0	0	
22	10003046	26048429	64	9.5	1.3	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
23	10003400	20214994	72	4.3	4.2	39	1.4	4	4	3.6	7.6	3.9	7.8	0	0	0	0	0	1.182692	98.26909	0	13.33333	16.4	0	0	
24	10003400	23559586	72	6.6	1.7	103	0.6	4	4	3.6	8.7	3.5	8.7	0	0	0	0	0	0.6875	23.15061	0	19.04762	21.4	1	0	
25	10003637	28317408	57	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	1	0	0	0	0	12.37624	0.6875	65.98807	0	18.66667	16	1	0
26	10004235	24181354	47	18.4	2	13	6.5	4	4	3.6	7.5	4.6	11.3	1	1	0	0	0	1372.581	0.991379	10.74787	1	16.14035	23.7	0	0
27	10004401	22869093	82	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
28	10004401	23920883	82	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.648148	65.98807	0	18.66667	16	0	0	
29	10004401	25777141	82	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
30	10004401	26488315	82	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.679612	65.98807	0	18.66667	16	0	0	
31	10004401	27939719	82	8.1	1.5	105	0.6	4	4	3.6	8.3	1.8	7.3	0	0	0	0	0	0.6875	64.08652	1	44.54545	11	0	0	
32	10004401	28128182	82	10.8	1.2	271	0.6	4	4	3.1	8.7	3.5	9.9	0	0	0	0	0	0.6875	80.78635	0	56.66667	13.2	0	0	
33	10004401	29988601	82	10.8	1.7	157	0.6	4	4	3.6	8.7	5	9.5	0	0	0	0	0	0.6875	41.59809	1	38.125	16.6	0	0	
34	10004422	21255400	78	12.8	0.8	193	0.6	4	4	3.6	9.2	3.8	13.4	0	0	0	0	0	0.6875	81.61068	0	32.22222	13.4	0	0	
35	10004457	23251352	65	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
36	10004696	29242151	64	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
37	10004720	22081550	61	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	
38	10004733	27411876	51	9.5	1.7	206	0.6	4	4	3.6	8.7	3.5	10.8	0	0	0	0	0	0.6875	65.98807	0	18.66667	16	0	0	

Figure 1: Cohort Construction from MIMIC-IV Dataset

After cleaning and extracting the data, we created the final dataset with 74,526 ICU stays, as shown in Figure 1, which displays a sample of the constructed cohort from the MIMIC-IV dataset.

- Feature Engineering & Selection** : The next crucial step involved selecting relevant features based on clinical expertise and data availability. Features were classified into the following categories:

Vitals (24h Pre-ICU): Heart rate, respiratory rate, blood pressure (systolic, diastolic), oxygen saturation, temperature, Glasgow Coma Scale (GCS).

Lab Results: Creatinine, BUN (Blood Urea Nitrogen), lactate, WBC (White Blood Cell) count, troponin, NT-proBNP, among others.

Derived Vitals & Scores: Mean Arterial Pressure (MAP), Shock Index, Quick SOFA score (qSOFA), and others.

Medications & Treatments: Including the use of vasopressors, antibiotics, and diuretics.

Outcomes (Target Variables): Sepsis, Heart Failure, Chronic Kidney Disease.

In total, we initially extracted 150+ features, but after preprocessing (imputing missing data, handling outliers, etc.), the final dataset included 41 key features that were deemed clinically relevant for predicting each condition.

- Model Development:** Three different predictive models were built for each disease (Sepsis, Heart Failure, and CKD). The models followed a structured process that included addressing class imbalances, applying class weights, and experimenting with multiple machine learning algorithms. Below is a summary of the methodology followed:

- Exclusion Logic per Model :**

Model 1 (Sepsis): Exclude Heart Failure and CKD labels during training.

Model 2 (Heart Failure): Exclude Sepsis and CKD labels.

Model 3 (CKD): Exclude Sepsis and Heart Failure labels.

- Class Imbalance Handling:** The dataset was imbalanced, especially for Heart Failure (only 6.5% of patients) and CKD (36%). Various techniques such as undersampling, class weighting, and hybrid models were employed to tackle this issue.

- Evaluation Metrics:** Accuracy: Overall performance metric. Precision/Recall: For clinical risk sensitivity and false-positive control. F1-Score: To balance between recall and precision. ROC-AUC: To assess the model's ability to rank classes properly.

- Training and Testing:** Models were trained using Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. Each model was evaluated on the balanced dataset (using undersampling) as well as the original dataset (with class weights applied to handle class imbalance).

Table 1: Model Performance Metrics on Balanced Dataset (Undersampling)

2*Model	Sepsis					Heart Failure					CKD				
	Acc	Prec	Recall	F1	AUC	Acc	Prec	Recall	F1	AUC	Acc	Prec	Recall	F1	AUC
XGBoost	0.620	0.601	0.531	0.564	0.664	0.650	0.479	0.293	0.364	0.674	0.602	0.547	0.439	0.487	0.649
Random Forest	0.628	0.600	0.592	0.596	0.658	0.639	0.469	0.439	0.454	0.656	0.602	0.537	0.551	0.544	0.677
Logistic Regression	0.598	0.574	0.507	0.539	0.656	0.626	0.470	0.758	0.580	0.688	0.683	0.686	0.485	0.568	0.696
Gradient Boosting	0.591	0.565	0.507	0.535	0.655	0.672	0.539	0.261	0.352	0.676	0.654	0.689	0.359	0.472	0.655

Table 2: Model Performance Metrics on Original Dataset (Class-Weighted)

2*Model	Sepsis					Heart Failure					CKD				
	Acc	Prec	Recall	F1	AUC	Acc	Prec	Recall	F1	AUC	Acc	Prec	Recall	F1	AUC
XGBoost	0.611	0.589	0.531	0.558	0.665	0.674	0.544	0.274	0.364	0.689	0.777	0.130	0.498	0.207	0.722
Random Forest	0.598	0.562	0.596	0.579	0.668	0.676	0.521	0.624	0.568	0.713	0.786	0.088	0.285	0.134	0.649
Logistic Regression	0.602	0.579	0.516	0.546	0.656	0.628	0.472	0.764	0.584	0.690	0.789	0.140	0.510	0.220	0.739
Gradient Boosting	0.633	0.615	0.554	0.583	0.669	0.665	0.518	0.280	0.364	0.685	0.945	0.700	0.113	0.194	0.754

Table 3: Model Performance Metrics for Hybrid Voting Models (Ensemble of Best Class-Weighted Models)

2*Model	Sepsis					Heart Failure					CKD				
	Acc	Prec	Recall	F1	AUC	Acc	Prec	Recall	F1	AUC	Acc	Prec	Recall	F1	AUC
Hybrid Voting Model	0.869	0.287	0.278	0.283	0.704	0.971	0.040	0.076	0.053	0.617	0.921	0.322	0.315	0.318	0.757

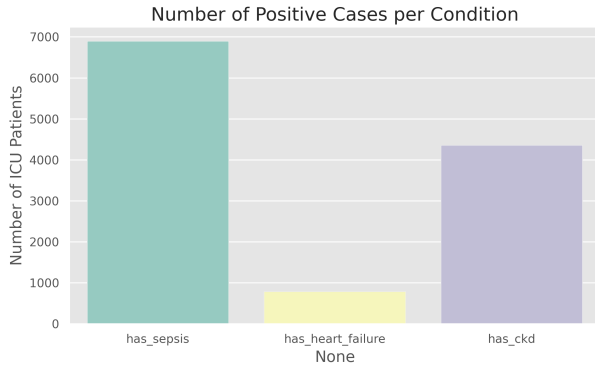


Figure 2: Number of ICU patients diagnosed with Sepsis, Heart Failure, and CKD.

Results: Model Performance Evaluation

The machine learning models evaluated for predicting Sepsis, Heart Failure (HF), and Chronic Kidney Disease (CKD) across three key phases (Balanced Dataset, Original Dataset with Class Weights, and Hybrid Models) revealed distinct performance patterns across the diseases. Below, we outline the performance across the three phases and key insights for each condition.

Phase 1: Balanced Dataset (Undersampling)

To address the significant class imbalance (with Heart Failure comprising only 6.5% of the dataset), we undersampled the Sepsis and CKD cases to match the size of the minority class (HF). This phase helped to better understand how each model performed when the class distribution was balanced.

Key Insights:

Sepsis: Gradient Boosting performed consistently well across all metrics, achieving the highest recall and ROC-AUC score, as shown in Table 1, indicating it is effective at identifying Sepsis cases even in the balanced data.

Heart Failure: Random Forest outperformed other models in terms of precision, recall, and F1-score, highlighting its strength in capturing true positives in Heart Failure prediction.

CKD: Random Forest showed balanced performance, excelling in accuracy and recall, crucial for early detection.

Phase 2: Original Dataset (Class-Weighted)

In this phase, we used the full dataset (74,526 ICU records), where class imbalance (57.3% Sepsis, 36.2% CKD, and 6.5% Heart Failure) was retained. As shown in Figure 2, the number of ICU patients diagnosed with each condition varied significantly. To address this imbalance, we applied class weights during model training.

Key Insights:

Sepsis: Gradient Boosting with class weighting demonstrated the highest recall and ROC-AUC, as shown in Table 2, confirming its ability to identify high-risk Sepsis patients despite class imbalance.

Heart Failure: Random Forest with class weighting continued to show the best balance between precision, recall, and F1-score, confirming its role as the optimal model for HF detection.

CKD: Random Forest continued to show solid performance, with its ability to generalize well to the minority class (CKD).

Phase 3: Hybrid Models (Voting + Stacking)

In this phase, ensemble methods were applied using soft voting and stacked generalization to combine the best base learners (Logistic Regression, Random Forest, Gradient Boosting, and XGBoost). These models aimed to improve decision-making flexibility and stability by combining multiple models' strengths.

Key Insights:

Sepsis: The Hybrid Voting model showed high accuracy but failed to capture precision and recall effectively as shown in Table 3. This suggests that while ensemble models can improve overall decision-making, they may sacrifice the sensitivity needed in clinical applications.

Heart Failure: The Hybrid model showed extremely high accuracy, but the recall and precision suffered, making it unsuitable for clinical use where identifying true positive cases is critical.

CKD: Hybrid models struggled with maintaining precision and recall, highlighting that combining multiple models in a voting mechanism does not necessarily improve performance for rare conditions like CKD.

Final Model Selection

Based on performance across all phases, the best models for each disease were selected. Gradient Boosting was chosen for Sepsis, while Random Forest was selected for both Heart Failure and CKD. These selections, shown in figure 7 and table 4, reflect each model's ability to consistently balance accuracy, precision, recall, and ROC-AUC using class-weighted training.

Learning Curves for Best Models per Disease

Learning curves provide insights into how each model behaves with different training set sizes.

- 1. Sepsis (Gradient Boosting):** Shows steady improvement in training and validation accuracy, indicating that the model benefits from a larger training set.

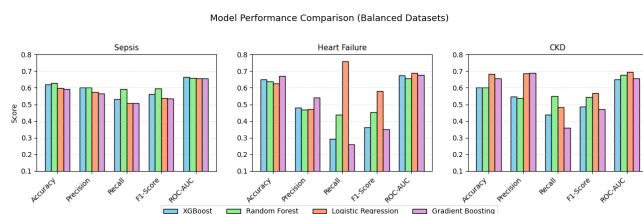


Figure 3: Model Performance on Balanced Dataset in phase 1

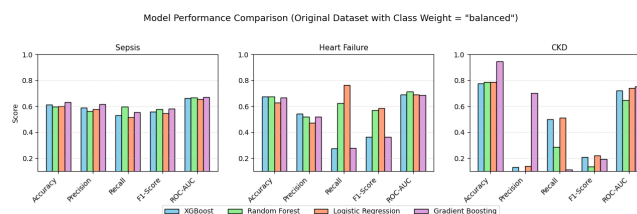


Figure 4: Model Performance on Original Dataset with Class Weights in phase 2

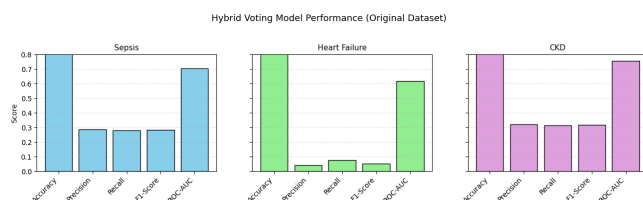


Figure 5: Hybrid Voting Model Performance in phase 3

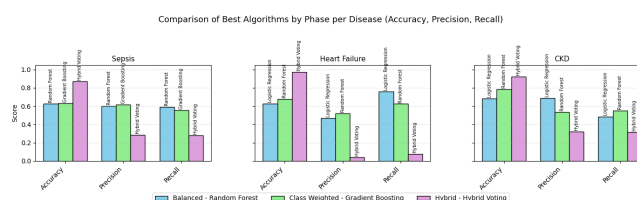


Figure 6: Best Performing Algorithms across phases

Table 4: Final selected models for Sepsis, Heart Failure, and CKD, chosen based on optimal performance across key metrics using class-weighted training

Disease	Selected Model	Model Type
Sepsis	Gradient Boosting	Class Weighted
Heart Failure	Random Forest	Class Weighted
CKD	Random Forest	Class Weighted

- Heart Failure (Random Forest):** Displays stable validation accuracy, confirming Random Forest's robustness in this setting.
- CKD (Random Forest):** Demonstrates consistent performance as training set size grows, with minimal overfitting observed.

Challenges and Lessons Learned

The process of building machine learning models for early disease prediction in ICU patients using the MIMIC-IV dataset presented several challenges. These challenges not only helped us gain deeper insights into machine learning in healthcare but also guided improvements and refinements during the project.

- Computation Overhead:** The MIMIC-IV dataset is vast, containing millions of records across multiple years of data. This resulted in significant computational overhead during data processing:

Data Size: The dataset consists of over 74,000 ICU admissions, with millions of data points from lab tests, vital signs, and medication records. Handling and processing such large data volumes required high computational power and optimized data structures to ensure that the workflow remained efficient.

Memory Optimization: Large data merges (combining ICU chart events, lab events, and clinical diagnoses)

placed a substantial load on system memory. To overcome this, parallel processing techniques were used to handle data chunks more efficiently, and unnecessary columns were dropped during the preprocessing phase.

Lesson Learned: It is crucial to ensure data optimization, particularly in healthcare datasets where data volume can grow exponentially. Efficient memory management and parallelization of tasks are key to handling large datasets without compromising performance.

- Class Imbalance :** One of the significant challenges in this project was the class imbalance present in the dataset. Specifically:

Heart Failure (6.5

The imbalance affected the performance of some machine learning models, especially in terms of precision and recall.

To address this:

Undersampling was applied to the minority classes, reducing the number of Sepsis and CKD cases to match the number of Heart Failure cases in the balanced dataset.

Class weighting was used in the models to ensure that rare diseases (e.g., Heart Failure) received more attention during training.

Lesson Learned: Class imbalance is a common issue in medical datasets, and while undersampling can help balance the data, class weighting is a more nuanced approach. It's essential to evaluate model performance

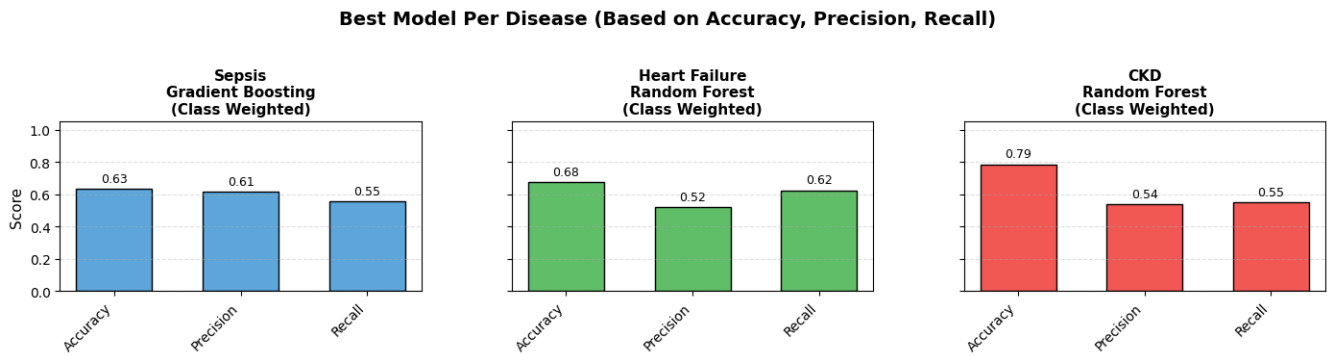


Figure 7: Final Selected best Models per Disease

across both balanced and imbalanced datasets to ensure real-world clinical effectiveness.

3. **Data Integration** : The MIMIC-IV dataset is distributed across several large tables, which include diverse data sources like chart events, lab events, and diagnoses. The integration of these datasets into a cohesive structure posed the following challenges:

Table Merging: Merging large, complex tables required careful handling to ensure no records were lost, and all relevant data points were retained. The merging process also had to account for time-aligning the data (e.g., ensuring that lab results and vitals corresponded to the same ICU stay).

Feature Validation: Ensuring the quality and clinical relevance of derived features like qSOFA and eGFR was critical. For instance, qSOFA was derived based on a set of vital signs, but the time window of 24 hours before ICU admission needed to be precisely aligned.

4. **Clinical Validity:** Ensuring that the features used for prediction were clinically valid was one of the most significant hurdles:

Derived Features: Features like qSOFA and eGFR are clinically relevant for detecting Sepsis and CKD, but their effectiveness relies on the correct clinical context and input values.

For example, qSOFA is a score based on vital signs, but a low qSOFA score may be acceptable for a patient with chronic health conditions. Thus, ensuring that features were interpreted correctly for each clinical case was essential.

Lesson Learned: It is essential to involve domain experts in clinical validation to ensure that the features used in the models are medically relevant and consistent with the clinical workflow.

Future Work

1. **UI Integration for Real-Time Use** : The current system can be extended into a real-time decision support system for ICU staff. This would allow for the prediction of critical conditions, with clear visual explanations using tools like SHAP. Integrating this into an interactive

dashboard would make it easier for clinicians to understand and trust model predictions while providing timely insights into patient conditions.

2. **Incorporating MIMIC-IV Waveform Data** : One potential extension involves incorporating high-frequency waveform data (e.g., ECG, SpO2, ABP) to detect early physiological anomalies before traditional clinical markers are observed. Adding waveform data could enhance the model's ability to make more accurate, dynamic predictions based on real-time patient status.
3. **Time-Series Deep Learning Models** : Given that the clinical data is time-dependent, applying Temporal Convolutional Networks (TCNs) or LSTMs (Long Short-Term Memory) could improve predictions by capturing temporal dependencies in the data. This could be particularly beneficial for modeling dynamic trends in patient vitals, enhancing the prediction of conditions like Sepsis, which can change rapidly over time.
4. **Multimodal Feature Fusion** : Another promising direction is to combine structured clinical data with unstructured data such as clinical notes, radiology reports, or discharge summaries. This would allow the model to capture a broader set of features and make better predictions by considering a variety of data sources.
5. **Clinician Feedback Loop** : Incorporating real-time clinician feedback into the model would help ensure that predictions remain aligned with clinical decision-making. This feedback loop could provide continuous model refinement, increasing trust in the system and improving its applicability in real-world clinical settings. Building on the insights gained during this project, several potential avenues for future work can be explored to improve model performance and expand the application of the system.

Conclusion

This project demonstrated the potential of interpretable machine learning models for early prediction of Sepsis, Heart Failure, and CKD in ICU patients. By leveraging structured data from the MIMIC-IV dataset and focusing on clinically relevant features, we developed models capable of providing actionable insights for clinicians. The Random Forest and Gradient Boosting models offered the best balance of accuracy and recall, prioritizing the detection of at-risk patients and minimizing false negatives.

The challenges faced such as class imbalance, data integration, and ensuring clinical validity were overcome using innovative methods like class weighting, undersampling, and SHAP analysis. Future work could focus on integrating real-time predictions, adding waveform data, and applying time-series models to enhance the system's accuracy and clinical utility.

By focusing on creating interpretable and clinically actionable tools, this project contributes to the ongoing effort to provide data-driven support for early intervention in critical care settings

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We also thank them for their guidance throughout the institutional approval process and for ensuring compliance with all ethical and regulatory standards. Their support was instrumental in enabling the successful completion of this project.

We are also grateful to the PhysioNet community for making the MIMIC-IV dataset publicly available under strict ethical protocols, fostering impactful research in the medical AI domain. Their efforts in data curation, maintenance, and documentation greatly facilitated our data preprocessing, cohort construction, and model training phases.

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Appendix A: Detailed Data Preparation, Feature Engineering, and Clinical Features

This appendix provides a detailed explanation of the data preparation, feature engineering, and clinical features used in this project. Each step was performed to ensure that the data was properly structured and preprocessed for machine learning model development. The following sections outline how we processed the MIMIC-IV dataset and derived key features for the model.

1. **Data Collection and Loading Core Tables** The MIMIC-IV v2.2 dataset was the primary data source for this project. The dataset contains various clinical tables, including patient demographics, ICU events, diagnoses, procedures, and supporting dictionaries. The process for loading and cleaning these core tables was as follows:

Loading Core Data Tables:

We loaded the following essential clinical tables using pandas:

- **patients:** Contains patient demographics and vital information.
- **admissions:** Includes hospital admission details.
- **transfers:** Tracks patient transfers between care units.
- **chartevents:** Contains real-time monitoring data, such as vital signs.
- **labevents:** Includes lab results like creatinine, sodium, and lactate.
- **inputevents, outputevents:** Track interventions like medications and fluids.
- **diagnoses_icd:** Contains patient diagnoses coded using ICD-10 codes.
- **procedures_icd:** Includes procedures performed on the patient during the ICU stay.

These tables were used to extract clinical and demographic features for the modeling process.

2. **Building the Adult ICU Cohort** To ensure the analysis focused on relevant ICU patients, the following steps were performed:

Filtering for Adult Patients:

We filtered for adult patients (≥ 18 years) using the `anchor_age` field from the `patients` table.

Identifying ICU Admissions:

The `transfers` table was used to identify and retain only those admissions that involved ICU units.

Excluding Short ICU Stays:

ICU stays less than 4 hours were excluded from the dataset, as these are not clinically relevant for this project.

Retaining First ICU Stay Per Patient:

For patients who had multiple ICU admissions, only the first ICU stay was kept to avoid bias from repeated ICU visits.

3. **Feature Engineering** Feature engineering was the critical step of selecting and transforming the relevant features from the clinical data to be used for modeling. Below are the features extracted:

Vital Signs

Several key vital signs were extracted from the `chartevents` table:

Heart Rate, Blood Pressure (SBP, DBP), Respiratory Rate, Oxygen Saturation (SpO₂), and Glasgow Coma Scale (GCS) were all included as important predictors for conditions like Sepsis and Heart Failure.

Formula for MAP:

$$MAP = \frac{2 \times DBP + SBP}{3}$$

Where:

SBP = Systolic Blood Pressure

DBP = Diastolic Blood Pressure

Lab Results

Key lab results were extracted from the `labevents` table. These included:

Creatinine: Used to estimate eGFR, an important measure for kidney function and CKD diagnosis.

Lactate: A critical marker for Sepsis and its severity.

WBC Count: Elevated in Sepsis and infection.

NT-proBNP and Troponin: Important biomarkers for Heart Failure.

Formula for eGFR:

$$eGFR = 175 \times (\text{Creatinine}^{-1.154}) \times (\text{Age}^{-0.203}) \times \begin{cases} 0.742 & \text{if female} \\ 1 & \text{if male} \end{cases} \quad (1)$$

Medications and Interventions

We also considered medication and intervention data, such as:

Vasopressors: Used to support blood pressure in Sepsis.

Antibiotics: Critical in the management of Sepsis.

Diuretics and ACE Inhibitors: Key treatments for Heart Failure.

A binary flag was created for each intervention (1 if the intervention occurred within the 24-hour pre-ICU window, 0 if it did not).

Clinical Scores

We computed several important clinical scores:

Shock Index: *Shock Index*:

$$\text{Shock Index} = \frac{\text{Heart Rate}}{\text{SBP}} \quad (2)$$

qSOFA Score:

$$\begin{aligned} qSOFA = & (\text{Respiratory Rate} \geq 22) \\ & + (\text{Systolic BP} \leq 100) \\ & + (\text{GCS} < 15) \end{aligned} \quad (3)$$

Note: A $qSOFA \geq 2$ is indicative of high Sepsis risk.

eGFR: As described above, this is a measure of kidney function critical for CKD diagnosis.

Outcome Labels

Outcome labels were defined using ICD-10 codes:

Sepsis: ICD-10 codes A40-A41.

Heart Failure: ICD-10 codes I50, I51.

CKD: ICD-10 codes N18, N19.

Each outcome was represented by a binary flag (1 for presence, 0 for absence).

4. **Final Dataset Construction** After completing the feature engineering process, we combined all the clinical features and outcome labels to create the final dataset. The dataset included:

Patient ID and Admission ID.

Clinical features such as vital signs, lab results, and clinical scores.

Binary outcome flags for Sepsis, Heart Failure, and CKD.

5. **Data Cleaning and Handling Missing Values** Before finalizing the dataset, we performed the following data cleaning steps:

Handling Missing Data:

Missing values for continuous variables (e.g., creatinine, lactate) were filled with the median value for that feature.

Binary flags (outcome labels and interventions) were filled with 0 where data was missing.